Unit IV
Design

Software Analysis & Design | CSIT 5th

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Data

- Data is a collection of a well defined small unit of information. It can be used in a variety of forms like text, numbers, media, bytes, etc. it can be stored in pieces of paper or electronic memory, etc.
- Word 'Data' is originated from the word 'datum' that means 'single piece of information.' It is plural of the word datum.
- In computing, Data is information that can be translated into a form for efficient movement and processing. Data is interchangeable.

Database

- Database is a collection of information (data & associated objects) which help to organize the related information in a logical fashion for easy access and retrieval.
- *Database, basically is nothing more than a computer based record-keeping system. The overall purpose of database is to record and maintain information required by organization in decision making.*

Database

- Database is a repository or collection of logically related, and similar data.
- Database stores similar kind of data that is organized in a manner that the information can be derived from it, modified, data added, or deleted to it, and used when needed.

- A database is defined as:

(1)a collection, or repository of data,

(2)having an organized structure, and

(3)for a specific purpose.

Database

A **database** is organized collection of related data of an organization stored in formatted way which is shared by multiple users.

The main feature of data in a database are:

1. It must be well organized
2. It is related.
3. It is accessible in a logical order without any difficulty.
4. It is stored only once.

E.g. University database maintains information about students, courses and grades in university.

Database

- ***Data*** that is stored more or less permanently in a computer, we term a database. The software that allows one or many persons to use and/or modify this data is a *database management system* (DBMS).
- A ***Database Management System*** *(DBMS)* is a collection of programs that enables an organization to store, modify and extract information from the databases.
- A ***Relational Data Base Management System*** (RDBMS) stores data in many related Tables. The user can ask complex questions from one or more of these related tables and answers come back to the user as ***Forms*** and **Reports**.

Database Management System:

- DBMS is a software system for creating, organizing and managing the database. DBMS handles all access to the database and manages the database. Managing the database implies that it provides a convenient environment to the user to perform operations on the database for creation, insertion, deletion, updating, and retrieval of data.
- DBMS defines the scope of the use of database. This keeps data secure from unauthorized access.
- The functionality of DBMS includes : (1) the database that contains interrelated data, and (2) a set of programs to access the data.
- The DBMS implements the three schema architecture - internal schema at internal level, conceptual schema at conceptual level, and external schema at external level.
- Some of the common commercial DBMSs are—Oracle Database, IBMs DB2, Microsoft's SQL Server and Microsoft Access. MySQL is a popular open source DBMS.

Database System:

- A database system integrates the collection, storage, and dissemination of data required for the different operations of an organization, under a single administration.
- A database system is a computerized record keeping system. The purpose of the database system is to maintain the data and to make the information available on demand.
- All organizations or enterprises have some basic common functions. They need to collect and store data, process data, and disseminate data for their various functions depending on the kind of organization. Some of the common functions include payroll, sales report etc.

Characteristics of Database Approach:

- **Data Redundancy is Minimized:** Database system keeps data at one place in the database. The data is integrated into a single, logical structure. Different applications refer to the data from the centrally controlled location. The storage of the data, centrally, minimizes data redundancy.
- **Data Inconsistency is Reduced:** Minimizing data redundancy using database system reduces data inconsistency too. Updating of data values becomes simple and there is no disagreement in the stored values.
- **Data is Shared:** Data sharing means sharing the same data among more than one user. Each user has access to the same data, though they may use it for different purposes. Authorized users are permitted to use the data from the database. Users are provided with views of the data to facilitate its use.

Characteristics of Database Approach:

- **Data Independence:** It is the separation of data description (metadata) from the application programs that use the data. Data descriptions are stored in a central location called the data dictionary. This property allows an organization's data to change and evolve (within limits) without changing the application programs that process the data.
- **Data Integrity maintain:** Stored data is changed frequently for variety of reasons such as adding new data item types, and changing the data formats. The integrity and consistency of the database are protected using constraints on values that data items can have.
- **Data Security improve:** The database is kept secure by limiting access to the database by authorized personnel. Authorized users are generally restricted to the particular data they can access, and whether they can update it or not. Access is often controlled by passwords.

Characteristics of Database Approach:

- **Backup and Recovery Support** : Backup and recovery are supported by the software that logs changes to the database. This support helps in recovering the current state of the database in case of system failure.
- **Standards are Enforced:** Since the data is stored centrally, it is easy to enforce standards on the database. Standards could include the naming conventions, and standard for updating, accessing and protecting data. Tools are available for developing and enforcing standards.
- **Application Development Time is Reduced:** The database approach greatly reduces the cost and time for developing new business applications. Programmer can focus on specific functions required for the new application, without having to worry about design, or low-level implementation details; as related data have already been designed and implemented. Tools for the generation of forms and reports are also available.

Data model Schema and Instance

- The data which is stored in the database at a particular moment of time is called an instance of the database.
- The overall design of a database is called schema.
- A database schema is the skeleton structure of the database. It represents the logical view of the entire database.
- A schema contains schema objects like table, foreign key, primary key, views, columns, data types, stored procedure, etc.
- A database schema can be represented by using the visual diagram. That diagram shows the database objects and relationship with each other.
- A database schema is designed by the database designers to help programmers whose software will interact with the database. The process of database creation is called data modeling.

Data model Schema and Instance

- A schema diagram can display only some aspects of a schema like the name of record type, data type, and constraints. Other aspects can't be specified through the schema diagram. For example, the given figure neither show the data type of each data item nor the relationship among various files.
- In the database, actual data changes quite frequently. For example, in the given figure, the database changes whenever we add a new grade or add a student. The data at a particular moment of time is called the instance of the database.

Data model Schema and Instance

STUDENT

Name	Student_number	Class	Major

COURSE

Course_name	Course_number	Credit_hours	Department

PREREQUISITE

Course_number	Prerequisite_number

SECTION

Section_identifier	Course_number	Semester	Year	Instructor

GRADE_REPORT

Student_number	Section_identifier	Grade

Approaches of Data Management

There are two approaches of data management:

1. File-oriented approach
2. Database approach

The *advantages of database system* over file system are:

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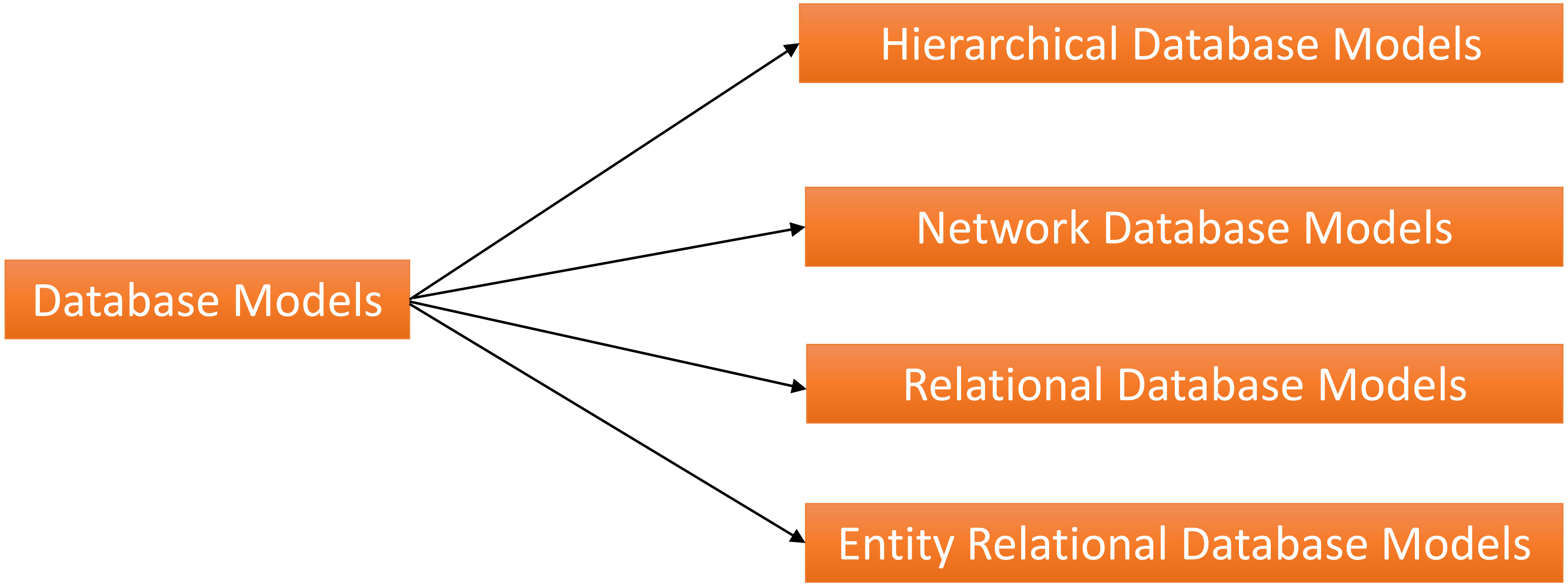
Disadvantage of DBMS

- Cost of Hardware and Software of a DBMS is quite high which increases the budget of your organization.
- Most database management systems are often complex systems, so the training for users to use the DBMS is required.
- In some organizations, all data is integrated into a single database which can be damaged because of electric failure or database is corrupted on the storage media
- Use of the same program at a time by many users sometimes lead to the loss of some data.
- DBMS can't perform sophisticated calculations

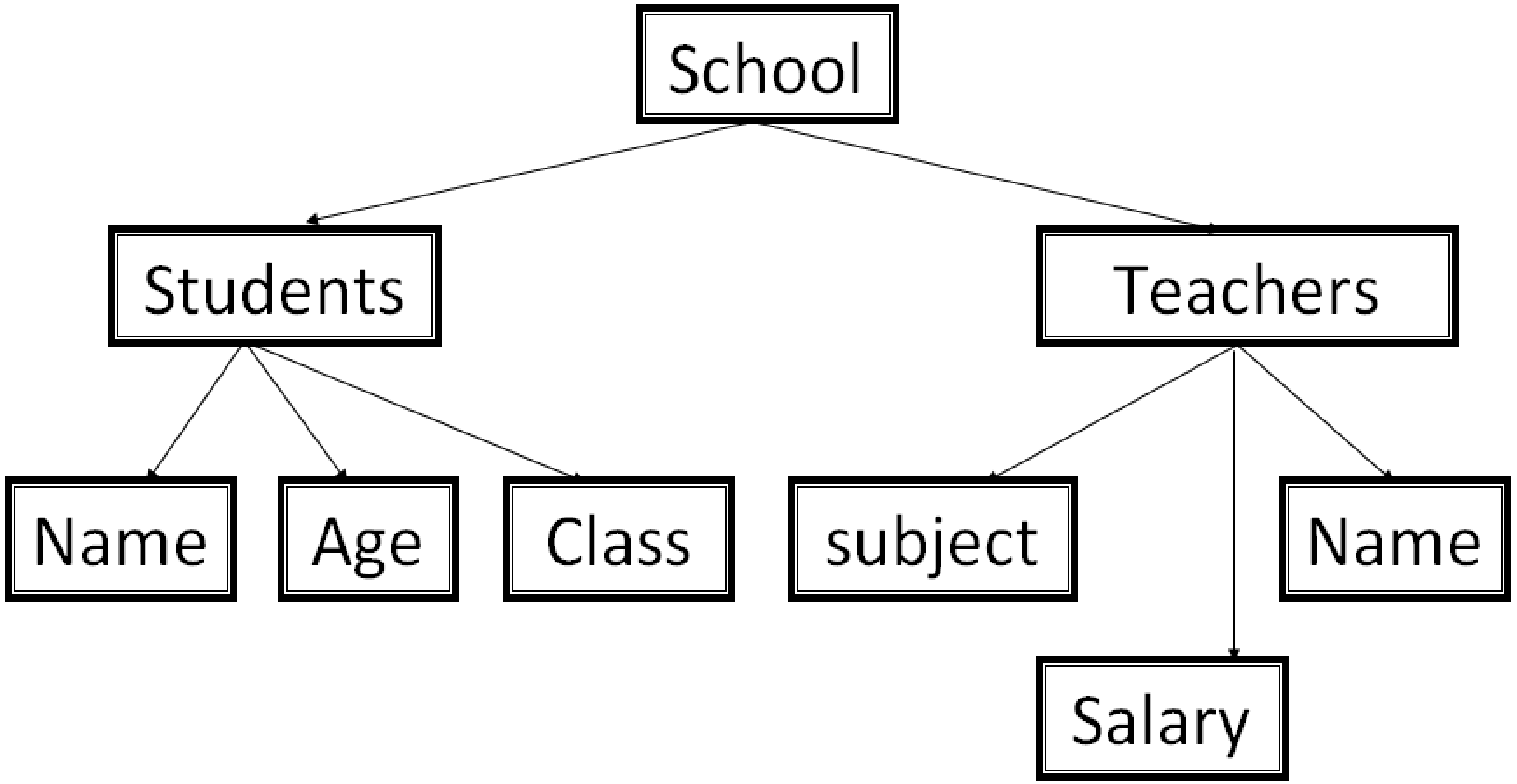
Database Models



Hierarchical Database Model

- The oldest type database models.
- This model is recognized as the first database model created by IBM in the 1960s.
- This database model organizes data into a tree-like-structure, with a single root, to which all the other data is linked. The hierarchy starts from the Root data, and expands like a tree, adding child nodes to the parent nodes. In this model, a child node will only have a single parent node. In this model, data is organized with one-to-one or one-to-many relationship.

Hierarchical Database Model



Hierarchical Database Model

Advantages

- It is the easiest model
- It has one or more attributes
- The searching is fast and easy, if parent is known.
- It supports one-to-one and one-to-many relationship.

Disadvantages

- It is old fashioned, outdated database model.
- It does not support many-to-many relationship.
- The dependency on parent node is not beneficial always.
- It increase redundancy because same data is to be repeated in different places.

Network Database Model

- This is an extension of the Hierarchical model. In this model data is organized in a graph structure rather than tree structure and are allowed to have more than one parent node.
- It consists of collection of records that are inter-related to each other with the help of relationship.
- Each record has multiple fields and each field has only one data value.
- This database model uses many-to-many relationships.

Network Database Model

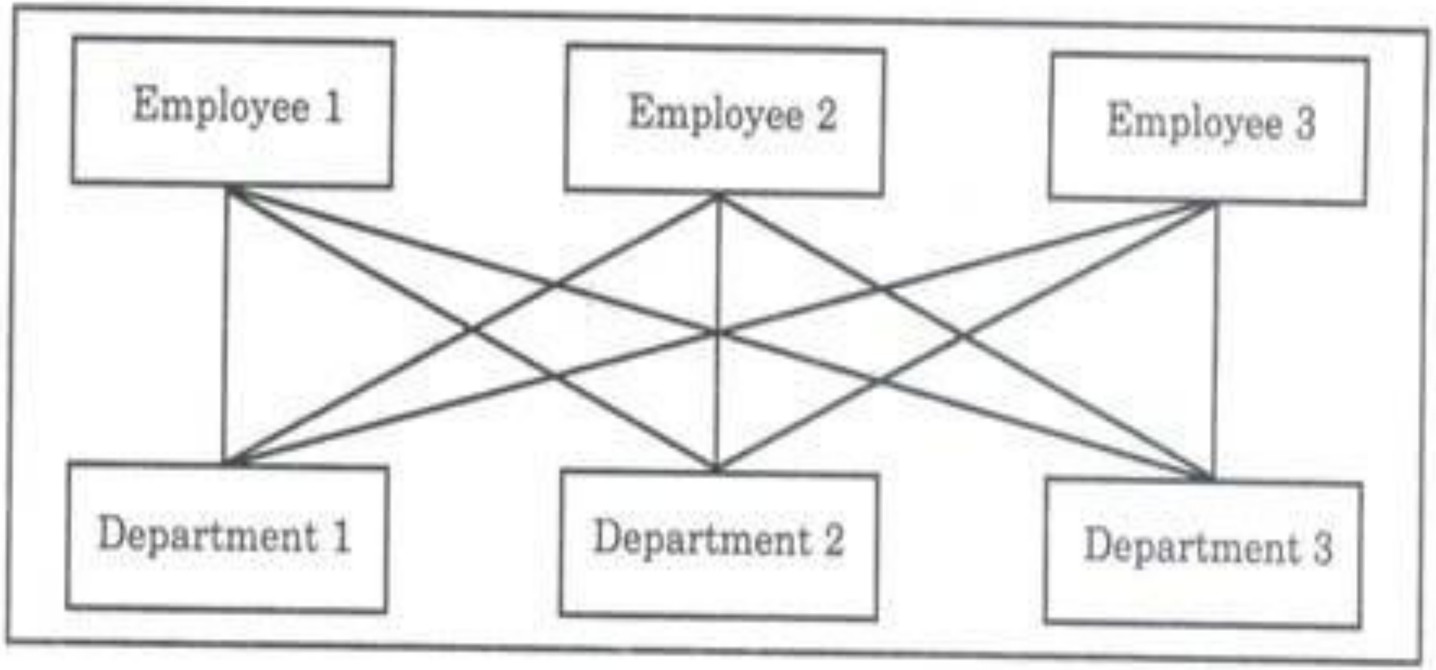


Fig. 9.5 Network model of employee information systems

Network Database Model

Advantages

- It accepts many-to-many relationship, so it is more flexible.
- The searching is faster because of multidirectional pointers.
- The network model is simple and easy to design.
- It reduces redundancy because data shouldn't be repeated if same data is needed.

Disadvantages

- It is difficult to handle the relationship in complex programs.
- There is less security because of sharing data.
- It increases the process overhead due to the complex relationship.

Relational Database Model

- RDBMS is the modern database approach.
- It uses a collection of tables to store data.
- Each table has unique name with rows and columns.
- A column defines the data field and a row defines a record in a table.
- Each data field must have atomic data value.
- Relationship is the association between two tables and it is represented by the common fields of both the tables.

There are four types of relationship in relational model:

one to one
one to many
many to one
many to many

Relational Database Model

Customer_detail			
Fname	**Lname**	**Street**	**Acc_no**
Abee	Gurung	Ratopool	A001
Rita	Shrestha	Lagankhel	A002
Ani	Lama	Jawalakhel	A003

Account_detail	
Acc_no	**Balance**
A001	54000
A002	38005
A003	97005

Relational Database Model

Advantages

- The breaking of complex database table into simple database table becomes possible.
- Database processing is faster than other model.
- There is very less redundancy.
- The integrity rules can easily be implemented

Disadvantages

- It is more complex than other models
- There are too many rules because of complex relationships.
- It needs more powerful computers and data storage devices.

Entity Relationship(ER) Model

An **Entity–relationship model (ER model)** describes the structure of a database with the help of a diagram, which is known as **Entity Relationship Diagram (ER Diagram)**. An ER model is a design or blueprint of a database that can later be implemented as a database. The main components of E-R model are: entity set and relationship set.

Main features of ER-model:

- Entity relationship model is a high level conceptual model
- It allows us to describe the data involved in a real world enterprise in terms of objects and their relationships.
- It is widely used to develop an initial design of a database
- It provides a set of useful concepts that make it convenient for a developer.

Entity-Relationship Database Model

- The diagrammatic representation of entities, attributes and relationship is called E-R diagram

Entity:

An entity is a “thing” or “Object” in the real world that is different from other objects.

Attribute:

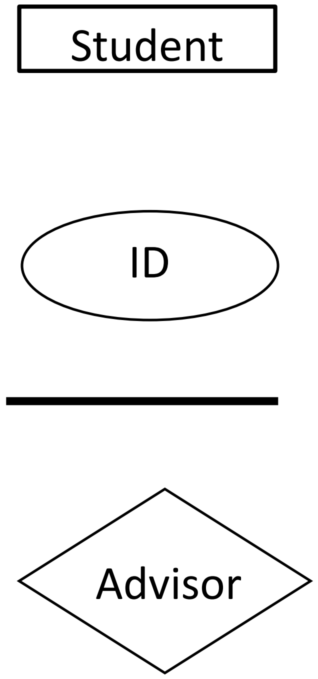
Properties possessed by an entity or relationship

Link:

Connector of entities, attributes and relationship

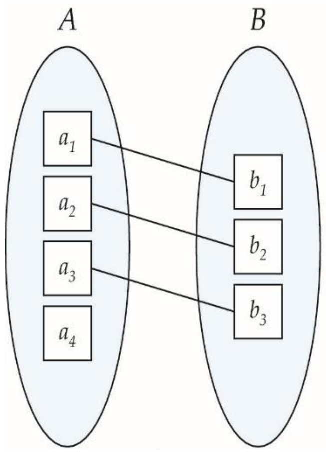
Relationship:

A relationship is an association among several entities and represents meaningful dependencies between them.

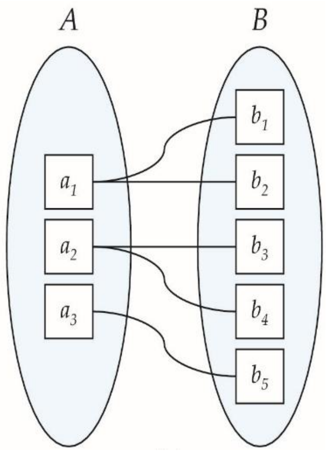


Entity-Relationship Database Model

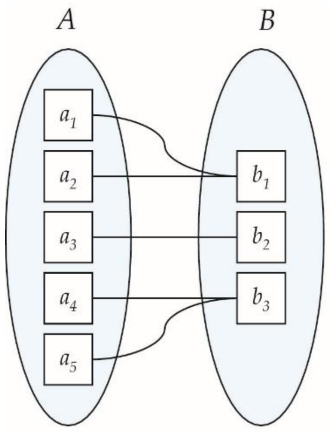
- Types of relationship:
  - One-to-one
  - One-to-many
  - Many-to-one
  - Many-to-many



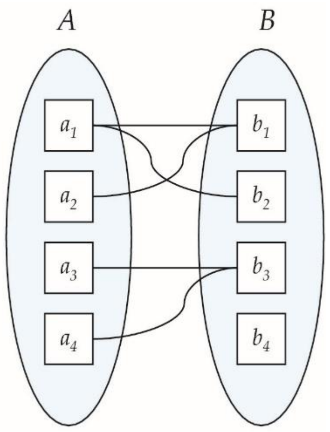
One to one



One to many

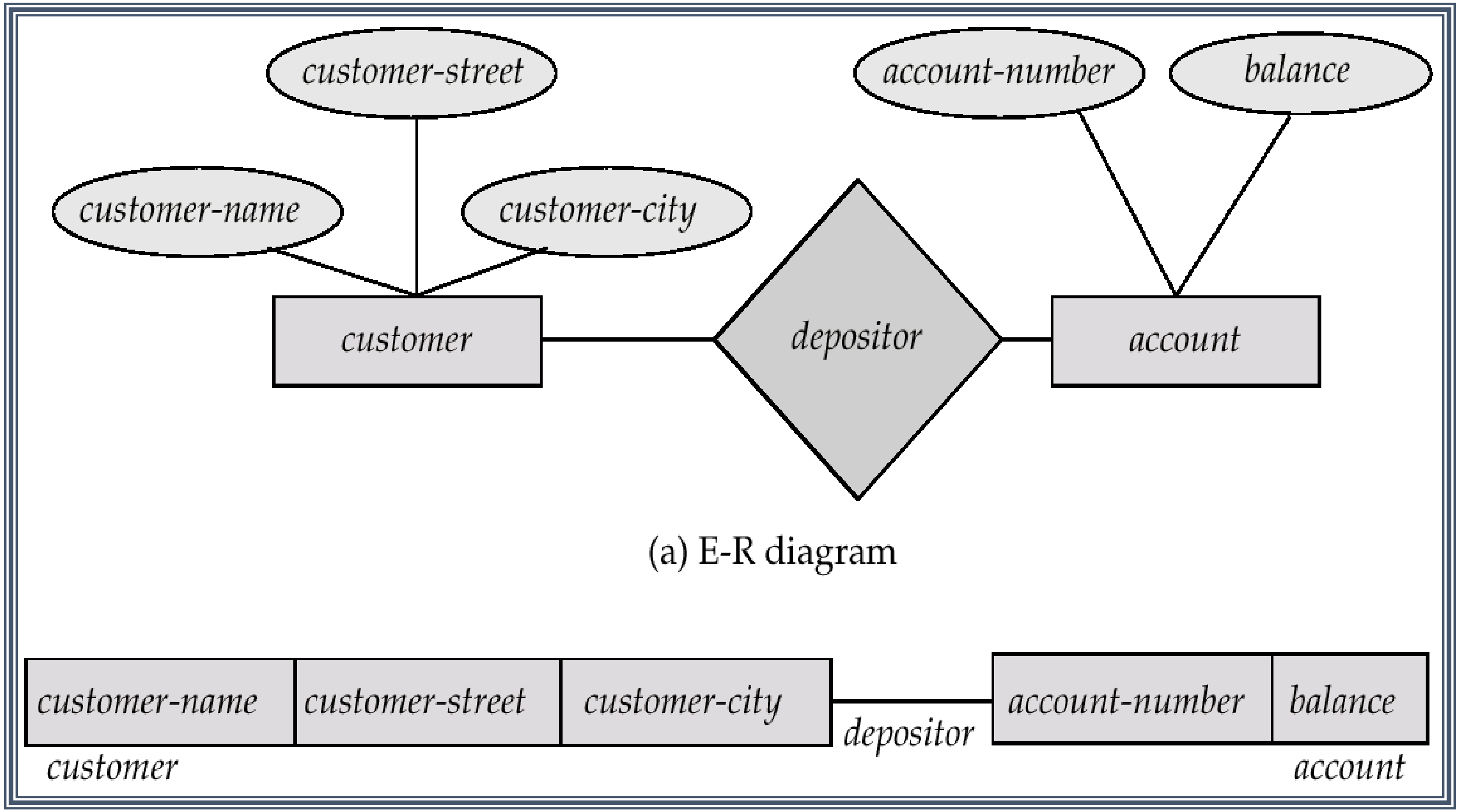


Many to one



Many to many

Entity-Relationship Database Model



Normalization

- It is a database design process in which complex database table is broken down into simple separate tables.
- It minimizes and controls the duplication of data in a database and also provides a rapid search for data from the database.
- It reduces the complexity of database.

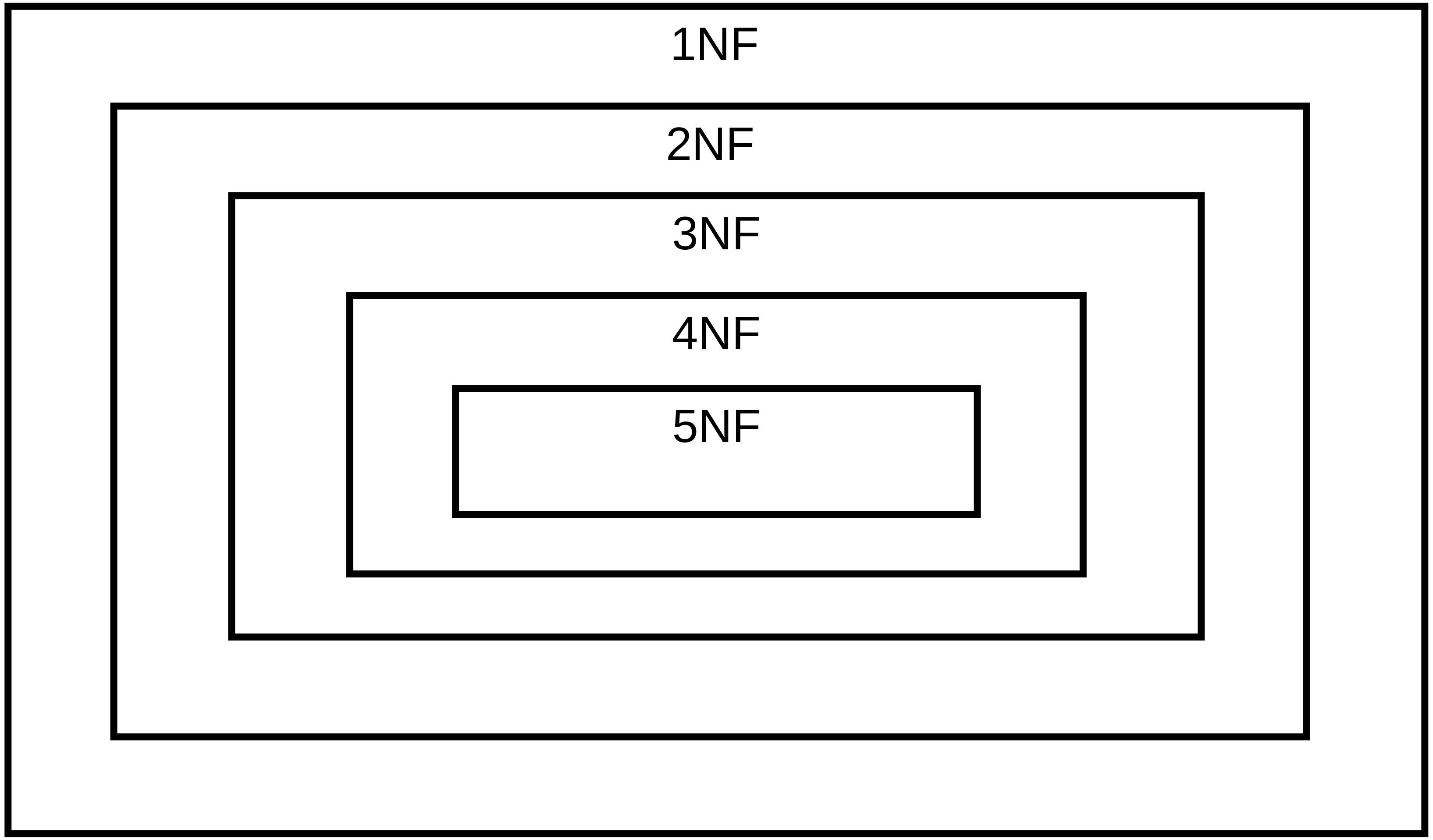
Advantages of Normalization

- The dependency between the data fields is identified.
- The redundancy in database is minimized.
- Data model is made more flexible and easier to maintain.
- It improves faster sorting and index creation.
- It improves the performance of the database system.
- It simplifies the structures of tables.
- It avoids the loss of information.

Levels of Normalization

- Various levels of normalization are:
  - First Normal Form (1NF)
  - Second Normal Form (2NF)
  - Third Normal Form (3NF)
  - BCNF
  - Fourth Normal Form (4NF)
  - Fifth Normal Form (5NF)

Levels of Normalization



First Normal Form (1NF)

- A relation is in first normal form if and only if the domain of each attribute contains only atomic (indivisible) values, and the value of each attribute contains only a single value from that domain.

1NF Example

Table: Student

Roll	Name	DOB	Marks
1	Abc	2056-10-15	410
2	Def	2057-04-08	415
3	Mno	2055-01-04	430
4	Xyz	2057-12-28	425

After First Normal Form

Roll	Name	Year	Month	Day	Marks
1	Abc	2056	10	15	410
2	Def	2057	04	08	415
3	Mno	2055	01	04	430
4	Xyz	2057	12	28	425

Second Normal Form (2NF)

- A table is in the second normal form if it is in the first normal form and it should not have Partial Dependency.
- All non-key attributes (columns) in the table should depend on the entire primary key, not just part of it.

2NF Example

Teacher_id	Subject	Teacher_age
111	Maths	38
222	Physics	38
222	Biology	38
333	Physics	40
333	Chemistry	40

After Second Normal Form

Table: **Teacher Detail**

Teacher_id	Teacher_age
111	38
222	38
333	40

Table: **Teacher Subject**

Teacher_id	Subject
111	Maths
111	Physics
222	Biology
333	Physics
333	Chemistry

Third Normal Form (3NF)

- A table is in the third normal form if it is in the second normal form and it doesn't have Transitive Dependency which means that non-key attributes should not depend on other non-key attributes..

3NF Example

Table: Vendor

ID	Name	AccountNo	BankCodeNo	Bank
A12345	Abc	143001743911	120	NABIL Bank
S30089	Def	109105043810	120	NABIL Bank
C33221	Mno	2330407152	845	Nepal Bank
F46785	Xyz	1451987450	845	Nepal Bank
D43780	Klm	110002042	777	Everest Bank
A43218	Pqr	427510254	777	Everest Bank

After Third Normal Form

3NF Example

Table: **Vendor**

ID	Name	AccountNo	BankCodeNo
A12345	Abc	143001743911	120
S30089	Def	109105043810	120
C33221	Mno	2330407152	845
F46785	Xyz	1451987450	845
D43780	Klm	110002042	777
A43218	Pqr	427510254	777

Table: **Bank**

BankCodeNo	Bank
120	NABIL Bank
845	Nepal Bank
777	Everest Bank

Database System:

- Components of Database System : A database system has four main components

  1.Users
    a) Application Programmers,
    b) End Users, and
    c) Data Administrators.

  2.Hardware

  3.Software
    a) Database Management System (DBMS),
    b) Application software, and
    c) User Interface.

  4.Data

Components of Database System

1.Users:

The users are the people who control and manage the databases and perform different types of operations on the databases in the database management system.

1. Application Programmers-The users who write the application programs in programming languages (such as Java, C++ or Visual Basic) to interact with databases are called Application Programmer.
2. End Users, -The end-users are those who interact with the database management system to perform different operations by using the different database commands such as Insert, update, retrieve and delete, on the data, etc. The end users interact with the system using query languages which requires some expertise for its use. End users include people like executives, managers, clerical staff, bank teller, and so on.
3. Data Administrators. A person who manages the overall DBMS is called a database administrator or simply DBA.

Components of Database System

2.Hardware:

- Hardware is the physical device on which the database system resides. The hardware required for the database system is the computer or a group of connected computers. Hardware also includes devices like magnetic disk drives, I/O device controllers, printers, tape drives, connecting cables and other auxiliary hardware.
- This component of DBMS consists of a set of physical electronic devices such as computers, I/O channels, storage devices, etc. that create an interface between computers and the users.
- This DBMS component is used for keeping and storing the data in the database.

Components of Database System

3.Software:

- The main component of a Database Management System is the software.
- In a database system, software lies between the stored data and the users of data. The database software can be broadly classified into three types…

1. Database Management System (DBMS),
2. Application software, and
3. User Interface.

- DBMS handles requests from the users to access the database for addition, deletion, or retrieval of data. Software components required to design the DBMS are also included here. Some software components are the automated tools like Computer-Aided Software Engineering (CASE) tools for the designing of databases and application programs, software utilities, and report writers.
- Application software uses DBMS facilities to manipulate the database to achieve a specific business function, such as, providing reports that can be used by users. Application software is generally written in a programming language like C, or it may be written in a language supported by the DBMS.
- User interface of the database system constitutes the menus and the screens that the user uses to interact with the application programs and the DBMS. It is the set of programs which is used to manage the database and to control the overall computerized database.
- The DBMS software provides an easy-to-use interface to store, retrieve, and update data in the database.

Components of Database System

4. Data :

- It is the most important component of the database management system.
- The main task of DBMS is to process the data. Here, databases are defined, constructed, and then data is stored, retrieved, and updated to and from the databases.
- The database contains both the metadata (description about data or data about data) and the actual (or operational) data.

Integrity Constraint

- Integrity Constraints are used to ensure accuracy and consistency of the data in a relational database. Integrity Constraints are set of rules that the database is not permitted to violate.
- Constraints may apply to each attribute or the may apply to relationships between tables.
- Integrity constraints ensure that changed (update, delete, insert) made to database by authorized users do not result in a loss of data consistency. Thus, integrity constraints guard against accidental damage to the database.

Types of Integrity Constraint

1. Domain Constraint
2. Entity Integrity Constraint
3. Referential Integrity Constraint
4. Key Constraints

1. Domain Constraints

- Domain Constraint defines the domain or the valid set of values for an attribute.
- The data type of domain includes string, character, integer, time, date, currency etc. The value of the attribute must be available in the corresponding domain.

STUDENT_ID	NAME	SEMESTER	AGE
101	Manish	1st	18
102	Rohit	3rd	19
103	Badal	5th	20
104	Amit	7th	A

Not allowed because age is an integer value

2. Entity Integrity Constraints

- The entity integrity Constraint states that primary key value can’t be null.
- This is because the primary key values is used to identify individual rows in relation and if the primary key has a null value, then we can’t identify those rows.
- A table can contain a null value other than the primary key field.

EMP_ID	EMP_NAME	SALARY
111	Mohan	20000
112	Rohan	30000
113	Sohan	35000
	Logan	20000

Not allowed as Primary Key can’t contain null value

3. Referential Integrity Constraints

- A referential integrity constraint is specific between two tables.
- Referential Integrity Constraint is enforced when a foreign key references the primary key of a table.
- In the referential integrity constraints, if a foreign key in table 1 refers to the primary key of table 2, every values of foreign key in table 1 must be available in primary key value of table 2 or it must be null.
- A table can contain a null value other than the primary key field.

3. Referential Integrity Constraints

Foreign Key

EMP (Table 1): Related /Referencing Table

EMP_ID	EMP_NAME	AGE	DEP_NO
111	Mohan	21	1
112	Rohan	33	2
113	Sohan	27	3
114	Logan	25	5

Relationships

Not allowed as DEP_NO 5 is not defined as a Primary Key of Table 2 and In Table 1, DEP_NO is a foreign key defined

Primary Key

DEP (Table 2): Primary/Referenced Table

DEP_NO	LOCATION
1	Mumbai
2	Delhi
3	Noida

4. Key Constraints

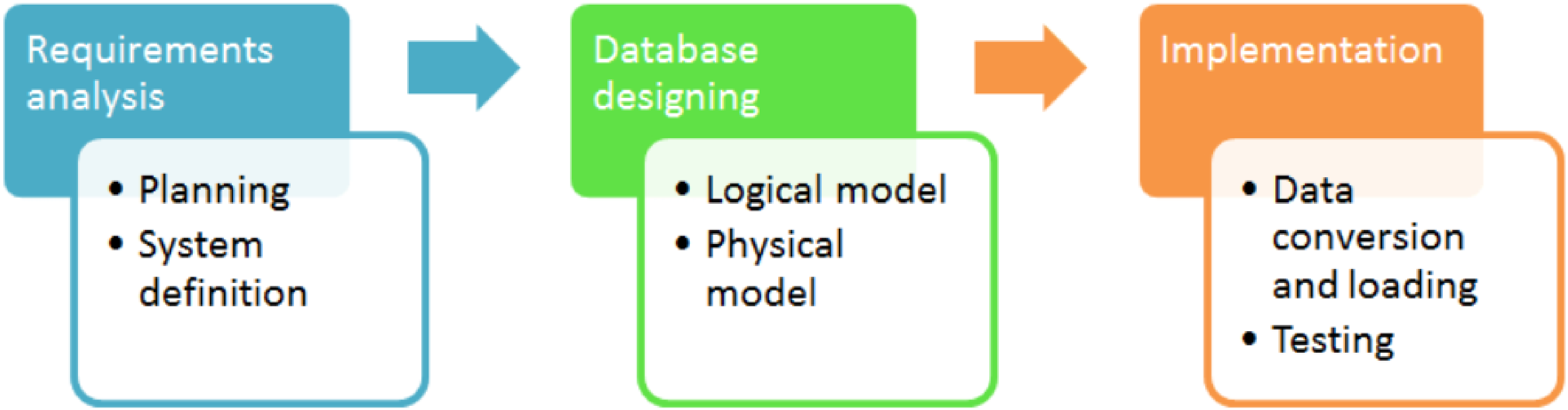
- An entity set can have multiple keys or candidate keys (minimal superkey), but out of which one key will be the primary key.
- Key constraint specifies that in any relation-
  - ❑ All the values of primary key must be unique.
  - ❑ The value of primary key must not be null.

STUDENT_ID	NAME	SEMESTER	AGE
101	Manish	1st	18
102	Rohit	3rd	19
103	Badal	5th	20
102	Amit	7th	21

Not allowed because all rows must be unique.

Designing Database

Database development life cycle



Requirements analysis

- **Planning -** This stages of database design concepts are concerned with planning of entire Database Development Life Cycle. It takes into consideration the Information Systems strategy of the organization.

- **System definition -** This stage defines the scope and boundaries of the proposed database system.

Database designing

•**Logical model -** This stage is concerned with developing a database model based on requirements. The entire design is on paper without any physical implementations or specific DBMS considerations.

•**Physical model -** This stage implements the logical model of the database taking into account the DBMS and physical implementation factors.

Implementation

•**Data conversion and loading** - This stage of relational databases design is concerned with importing and converting data from the old system into the new database.

•**Testing** - This stage is concerned with the identification of errors in the newly implemented system. It checks the database against requirement specifications.

Designing Database

- Logical database design
- Physical database design

Logical Database Design

- The logical database Model is concerned with collecting information about the business needs. It doesn’t focus on database design. The information that the logical database Model gathers are about business entities, organizational units, and business processes. It is mainly represented via diagrams. Its structures are abstract, which shows the domain of information. It is very much simpler than the physical database Model. Once the information is compiled, reports and diagrams are made, including these:
- **ERD**–Entity relationship diagram shows the relationship between different categories of data and shows the different categories of data required for the development of a database.
- **Business process diagram**–It shows the activities of individuals within the company. It shows how the data moves within the organization based on which application interface can be designed.

Logical Database Design

- Four key steps in logical database modeling and design:
  1. Develop a logical data model for each known user interface for the application using normalization principles.
  2. Combine normalized data requirements from all user interfaces into one consolidated logical database model (view integration).
  3. Translate the conceptual E-R data model for the application into normalized data requirements.
  4. Compare the consolidated logical database design with the translated E-R model and produce one final logical database model for the application.

Physical Database Design

- Physical database modeling deals with designing the actual database based on the requirements gathered during logical database modeling.
- All the information gathered is converted into relational models and business models. During physical modeling, objects are defined at a level called a schema level
- During physical design, you transform the entities into tables, the instances into rows, and the attributes into columns.

Physical Database Design

- After completing the logical design of your database, you now move to the physical design. You and your colleagues need to make many decisions that affect the physical design, some of which are listed below.
  - How to translate entities into physical tables
  - What attributes to use for columns of the physical tables
  - Which columns of the tables to define as keys
  - What indexes to define on the tables
  - What views to define on the tables
  - How to resolve many-to-many relationship

Relational Database Model

- **Relational database model**: data represented as a set of related tables or relations
- **Relation**: a named, two-dimensional table of data; each relation consists of a set of named columns and an arbitrary number of unnamed rows

RDBMS

- A database management system that stores data in the form of **related tables** is called Relational Database Management System.
- The goal of RDBMS is to make data easy to store & retrieve
- Relational databases help solve problems as they are designed to create tables & then combine the information in interesting ways to create valid information.

RDBMS

Relation Instance:- snapshot of DB

Example:

branch-name	*branch-city*	*assets*
Brighton	Brooklyn	7100000
Downtown	Brooklyn	9000000
Mianus	Horseneck	400000
North Town	Rye	3700000
Perryridge	Horseneck	1700000
Pownal	Bennington	300000
Redwood	Palo Alto	2100000
Round Hill	Horseneck	8000000

Schema :- Logical design of DB.

Example:

Account-schema = (account-number, branch-name, balance)

Branch-schema = (branch-name, branch-city, assets)

Customer-schema = (customer-name, customer-street, customer-city)

RDBMS

Schema Diagram

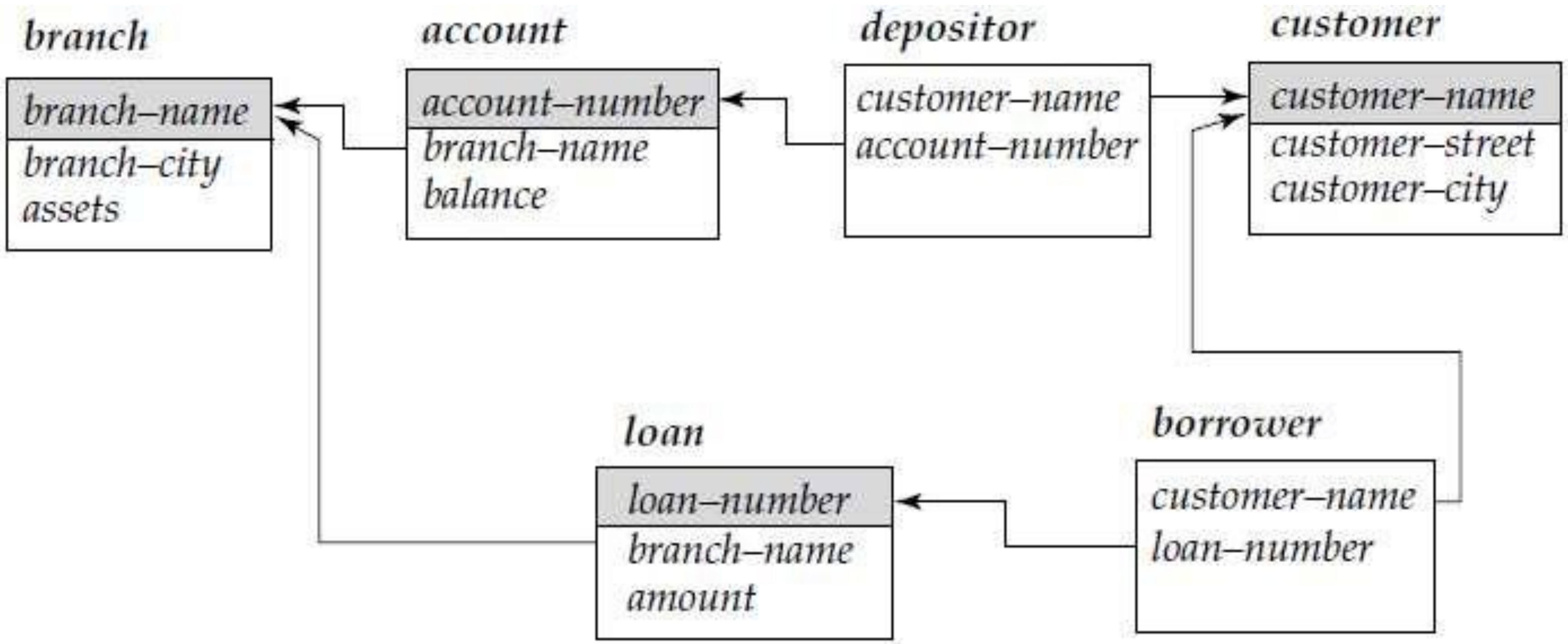
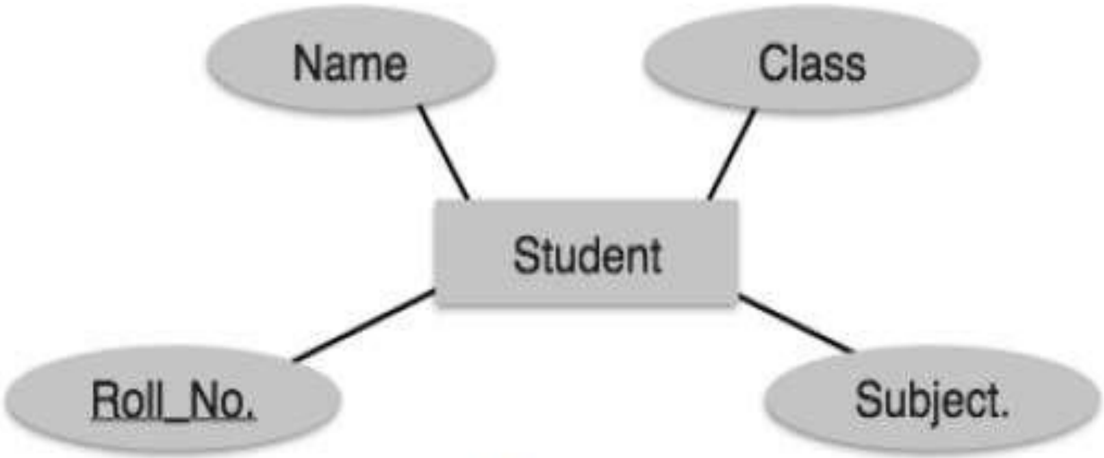


Figure 3.9 Schema diagram for the banking enterprise.

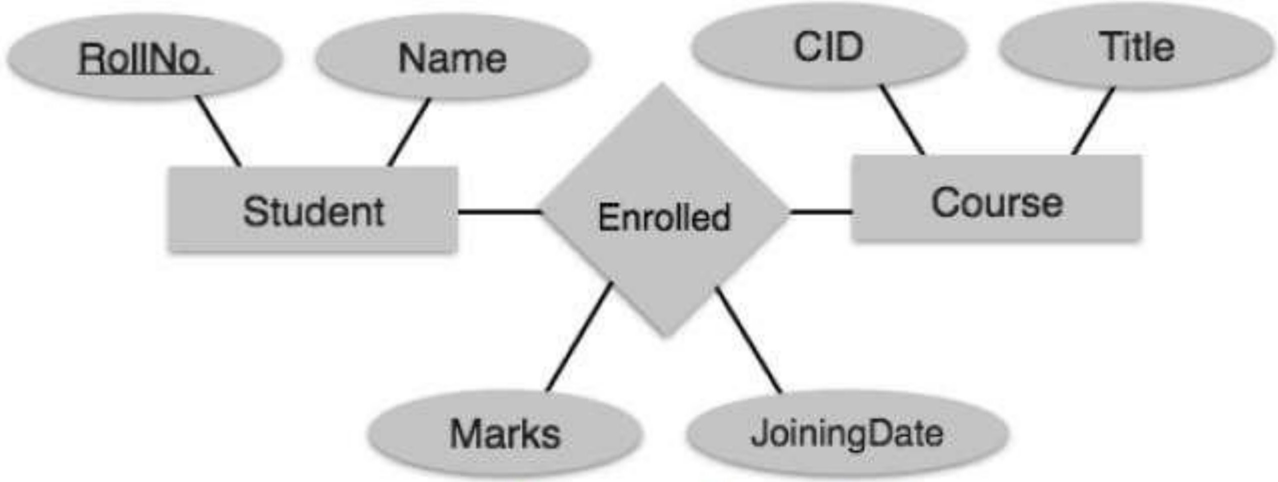
Translation ER Model to Relational Model



[Image: Mapping Entity]

- Create table for each entity
- Entity's attributes should become fields of tables with their respective data types.
- Declare primary key

Translation of ER Model to Relational Model



[Image: Mapping relationship]

- Create table for a relationship
- Add the primary keys of all participating Entities as fields of table with their respective data types.
- If relationship has any attribute, add each attribute as field of table.
- Declare a primary key composing all the primary keys of participating entities.
- Declare all foreign key constraints.

Merging Relations

- The normalization process tends to increase the number of relations by splitting unnormalized relations into smaller, normalized, relations. Conversely, decreasing the number of relations in a database by merging relations **reduces the need for joining relations**, and usually results in a better access performance.
- **View Integration**–Combining relations from multiple ER models into common relations
- Example :Employee1(EmployeeID, Name Address, Phone)

  Employee2(EmployeeID, Name, Address, Job Code, No. of Years)
- After merging

  Employee(EmployeeID, Name, Address, Phone, Job Code, No. of Years)

Physical File and Database Design

- The following information is required:
  - Normalized relations, including volume estimates
  - Definitions of each attribute
  - Descriptions of where and when data are used, entered, retrieved, deleted, and updated (including frequencies)
  - Expectations or requirements for response time and data integrity
  - Descriptions of the technologies used for implementing the files and database

Designing Fields

- **Field**: the smallest unit of named application data recognized by system software
  - Attributes from relations will be represented as fields
- **Data Type**: a coding scheme recognized by system software for representing organizational data

Choosing Data Types

- Selecting a data type balances four objectives:
  - Minimize storage space.
  - Represent all possible values of the field.
  - Improve data integrity of the field.
  - Support all data manipulations desired on the field.

TABLE 9-2 Commonly Used Data Types in Oracle 10i

Data Type	Description
VARCHAR2	Variable-length character data with a maximum length of 4000 characters; you must enter a maximum field length (e.g., VARCHAR2(30) for a field with a maximum length of 30 characters). A value less than 30 characters will consume only the required space.
CHAR	Fixed-length character data with a maximum length of 255 characters; default length is 1 character (e.g., CHAR(5) for a field with a fixed length of five characters, capable of holding a value from 0 to 5 characters long).
LONG	Capable of storing up to two gigabytes of one variable-length character data field (e.g., to hold a medical instruction or a customer comment).
NUMBER	Positive and negative numbers in the range 10^{-130} to 10^{126}; can specify the precision (total number of digits to the left and right of the decimal point) and the scale (the number of digits to the right of the decimal point) (e.g., NUMBER(5) specifies an integer field with a maximum of 5 digits and NUMBER(5, 2) specifies a field with no more than five digits and exactly two digits to the right of the decimal point).
DATE	Any date from January 1, 4712 B.C. to December 31, 4712 A.D.; date stores the century, year, month, day, hour, minute, and second.
BLOB	Binary large object, capable of storing up to four gigabytes of binary data (e.g., a photograph or sound clip).

Designing Forms and Reports

Introduction to Form Design

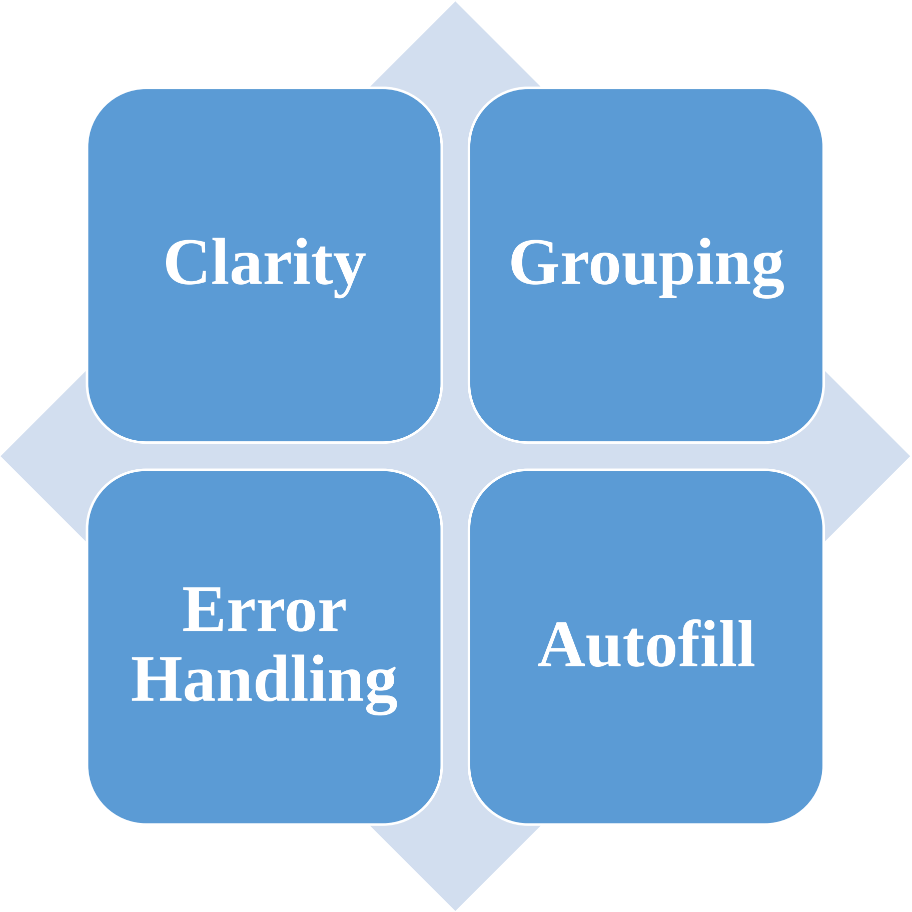
- **Definition**: A form is a graphical user interface (GUI) element that allows users to input data into a system.

- **Purpose**: Forms are used for data collection, registration, sign-up, and various interactive tasks.

- **Key Elements**: Labels, input fields, checkboxes, radio buttons, dropdowns, and submission buttons.

Best Practices for Form Design

- **Clarity**: Use clear and concise labels to indicate the purpose of each form field.
- **Grouping**: Group related fields together to improve readability and organization.
- **Error Handling**: Provide informative error messages to guide users in correcting form submissions.
- **Autofill**: Implement autofill features to assist users in filling repetitive information.



Introduction to Report Design

- **Definition**: A report is a structured presentation of data, analysis, or findings from a system or process.
- **Purpose**: Reports provide information, insights, and summaries for decision-making.
- **Key Elements**: Headings, subheadings, charts, graphs, tables, and visualizations.

Best Practices for Report Design

- **Clarity**: Present information in a clear and easy-to-understand manner.
- **Visuals**: Utilize charts and graphs to illustrate data trends and patterns effectively.
- **Consistency**: Maintain a consistent layout and style throughout the report.
- **White Space**: Use white space to enhance readability and emphasize key points.

Designing Forms and Reports Key Concepts

Form

- A business document that contains some predefined data and may include some areas where additional data are to be filled in
- An instance of a form is typically based on one database record

Report

- A business document that contains only predefined
- data A passive document for reading or viewing
- data

Typically contains data from many database records or transactions

Designing Forms and Reports

- Form design is the process of creating a web form — where the site visitors can input and submit their information — while keeping the form's layout, format, UX, appearance, and other factors in mind.
- The design of form impacts website's overall user experience (UX), which in turn directly impacts the number of happy visitors and conversions. A well-designed form shows users that your brand is helpful, thoughtful, professional, tech-savvy, and enjoyable to work with.
- A poorly-designed form, on the other hand, may lead to page and website abandonment or a frustrated user and, therefore, a decrease in conversions and sales.

Form Design Best Practices

- Be simple and straightforward. ...
- Use one column. ...
- Arrange form fields from easiest to hardest. ...
- Use inline form field validation. ...
- Align text to the left. ...
- Clearly title the form. ...
- Don't ask for phone numbers. ...

The Process of Designing Forms and Reports

- User-focused activity
- Follows a prototyping approach
- Requirements determination
  - Who will use the form or report?
  - What is the purpose of the form or report? When is the report needed or used?
  - Where does the form or report need to be delivered and used?
  - How many people need to use or view the form or report?

Prototyping

- Initial prototype is designed from requirements
- Users review prototype design and either accept the design or request changes
- If changes are requested, the construction- evaluation- refinement cycle is repeated until the design is accepted

General Formatting Guidelines for Forms and Reports

- ➢ **Highlighting**
- ➢ **Color versus No-Color**
- ➢ **Displaying Text**
- ➢ **Designing Tables and List**
- ➢ **Paper Versus Electronic Report**

Highlighting

- Blinking and audible tones should only be used to highlight critical information requiring user’s immediate attention.
- Methods should be consistently selected and used based upon level of importance of emphasized information

Color vs Non Color

- The coloring scheme should depend on the type of form.
- Color selection should be pleasant to eye.
- Choosing the right website color scheme can make your website more memorable, trustworthy, attractive and profitable.
- Labeling should be in dark color.
- Validation text should be in dark color.
- When building a site, user should consider carefully which colors to add to scheme and why. Each color will send different messages to the visitors and change the way they see the website.

- **Benefits from Using Color**
  - Soothes or strikes the eye
  - Accents an uninteresting display
  - Facilitates subtle discriminations in complex displays
  - Emphasizes the logical organization of information
  - Draws attention to warnings
  - Evokes more emotional reactions
- **Problems from Using Color**
  - Color pairings may wash out or cause problems for some users
  - Resolution may degrade with different displays
  - Color fidelity may degrade on different displays
  - Printing or conversion to other media may not easily translate

Displaying Text

- Display text in mixed upper and lower case and use conventional punctuation
- Use double spacing if space permits.
- Left-justify text and leave a ragged right margin. Do not hyphenate words between lines.
- Use abbreviations and acronyms only when they are widely understood by users and are significantly shorter than the full text

Designing tables and lists

Labels

- All columns and rows should have meaningful labels.
- Labels should be separated from other information by using highlighting.
- Re-display labels when the data extend beyond a single screen or page(page to page information)

Designing tables and lists (continued)

Formatting columns, rows and text

- Sort in a meaningful order
- Place a blank line between every five rows in long columns
- Similar information displayed in multiple columns should be sorted vertically
- Columns should have at least two spaces between them

  Allow white space on printed reports for user to write notes

  Use a single typeface, except for emphasis
- Use same family of typefaces within and across displays and reports
- Avoid overly fancy fonts

Designing tables and lists (continued)

Formatting numeric, textual and alphanumeric data

- Right-justify numeric data and align columns by decimal points or other delimiter
- Left-justify textual data. Use short line length, usually 30 to 40 characters per line
- Break long sequences of alphanumeric data into small groups of three to four characters each.

Designing Forms and Reports

Lightweight Graphics

- The use of small images to allow a Web page to be displayed more quickly

Forms and Data Integrity

- All forms that record information should be clearly labeled and provide room for input
- Clear examples of input should be provided to reduce data errors
- Site must clearly designate which fields are required, which are optional and which have a range of values

Paper versus Electronic Reports

- Printer used for producing paper report needs to be considered in design
- Use a prototyping process similar to designing a form

General guidelines for design of Forms and Reports

- **Meaningful Titles:**
  - Clear and specific titles describing content **and** use of form or report
  - Revision date or code to distinguish a form or report from prior versions
  - Current date, which identifies when the form or report was generated
  - Valid date, which identifies on what date (or time) the data in the form or report were accurate
- **Meaningful Information:**
  - Only **needed** information should be displayed
  - Information should be provided in a manner that is usable without modification
- **Balance the Layout:**
  - Information should be balanced on the screen or page
  - Adequate spacing **and** margins should be used
  - All data **and** entry fields should be clearly labeled
- **Design an Easy Navigation System:**
  - Clearly show how to move forward **and** backward
  - Clearly show where you are (e.g., page 1 of 3)
  - Notify user when on the last page of a multipage sequence

Assessing Usability

Overall evaluation of how a system performs in supporting a particular user for a particular task

Three characteristics

1. Speed
2. Accuracy
3. Satisfaction

Assessing Usability

- **Success Factors**
  - **Consistency**
    - Design elements all appear in same place on all forms forma and reports
    - Table presents usability factors and associated guidelines.
  - **Context**
    - Users
    - Tasks
    - Environment
    - Table present several characteristics that may influence the usability of a design

Assessing Usability

- **Measures of Usability**
  - Considerations
    - Time to learn
    - Speed of performance
    - Rate of errors
    - Retention over time
    - Subjective satisfaction
  - Collection Methods
    - Observation
    - Interviews
    - Keystroke Capturing
    - Questionnaires

Guidelines for Maximizing Usability

- Consistency: Consistency of terminology, formatting, titles, navigation, response time
- Efficiency: minimize required user actions
- Ease: self-explanatory outputs and labels
- Format: appropriate display of data and symbols
- Flexibility: maximize user options for data input according to preference

Designing Forms and Reports

Template-based HTML

- Templates to display and process common attributes of higher-level, more abstract items
- Creates an interface that is very easy to maintain

Designing Interfaces and Dialogues

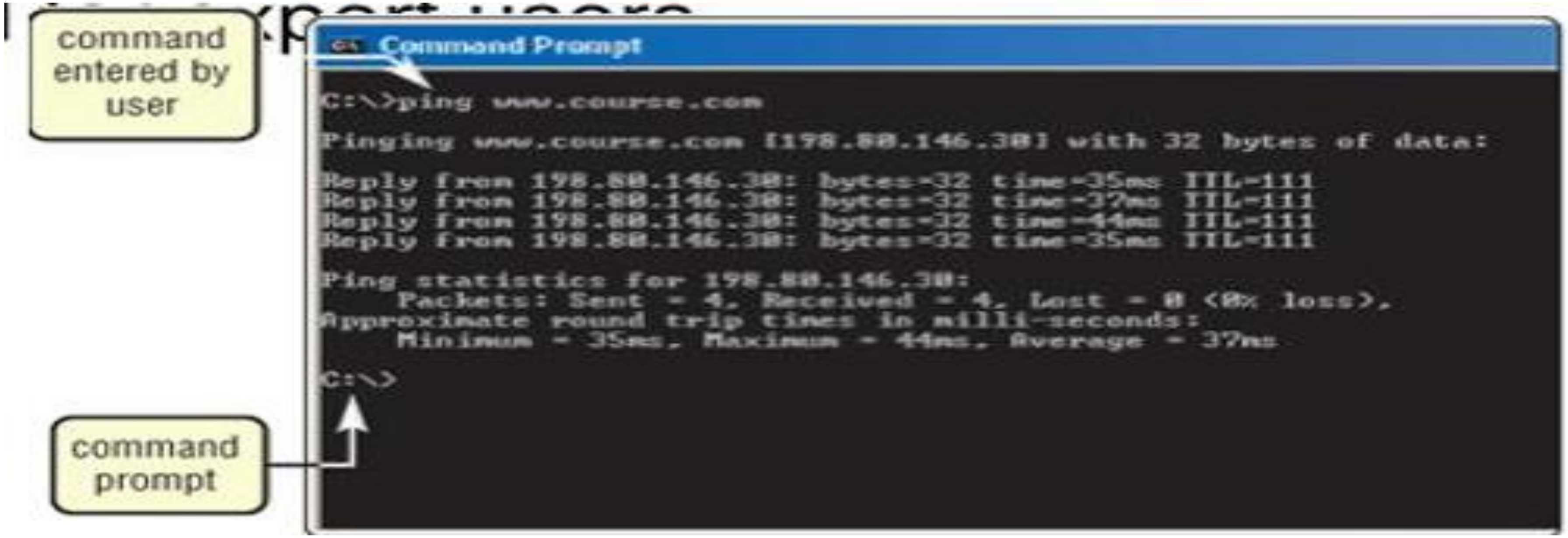
- A dialogue reflects the sequence of interaction between a user and a system. An interface is a method by which users interact with information systems.
- The process of designing interfaces and dialogues is similar to that of designing forms and reports . Thus, the design process is a user-focused activity that typically follows a prototyping approach.
- First, you must gain an understanding of the intended user and task objectives by collecting initial requirements during requirements determination. After collecting the initial requirements, you structure and refine this information into an initial prototype.

Interaction methods & Devices

- Interface: a method by which users interface with an information system.
- Methods of Interaction for designing usable interfaces include:
  1. **Command Language Interaction**
  2. **Menu Interaction (Pop-Up & Drop-Down)**
  3. **Form Interaction**
  4. **Object Based Interaction**
  5. **Natural Language Interaction**

1. Command Language Interaction

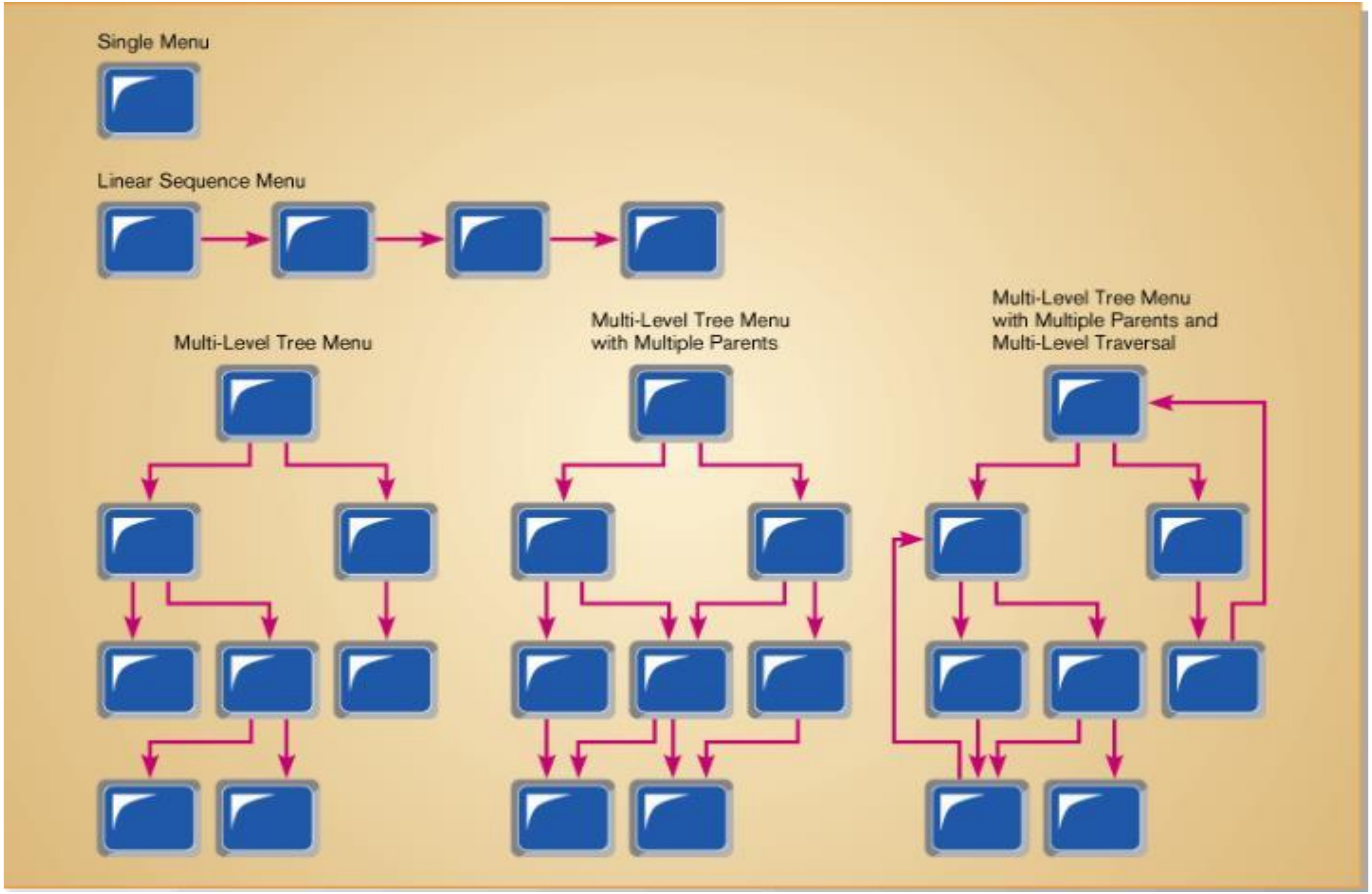
- Users enter explicit statements into a system to invoke operations within a system
- Difficult to interact with
  - Requires remembering commands, names & syntax.



2. Menu Interaction (Pop-Up & Drop-Down)

- A method in which a list of options is provided and specific command is invoked by user selection of a menu option.
- Menu complexity varies according to the needs of a system and capabilities of development environment.
  - **Single Menu**
  - **Hierarchies Levels**
- Two common placement methods:
  i. Pop-up
  ii. Drop-down

Figure 12-4 Single-level menu (along left edge of screen) from SnagIt® by TechSmith.


Single Menu
Linear Sequence Menu
Multi-Level Tree Menu
Multi-Level Tree Menu with Multiple Parents
Multi-Level Tree Menu with Multiple Parents and Multi-Level Traversal

i. Pop-up Menu

- A menu positioning method that places a menu near the current cursor position
- Used to:
  - Group commands for a certain job
  - Provide a list of possible use.
  - Fill a table with valid values

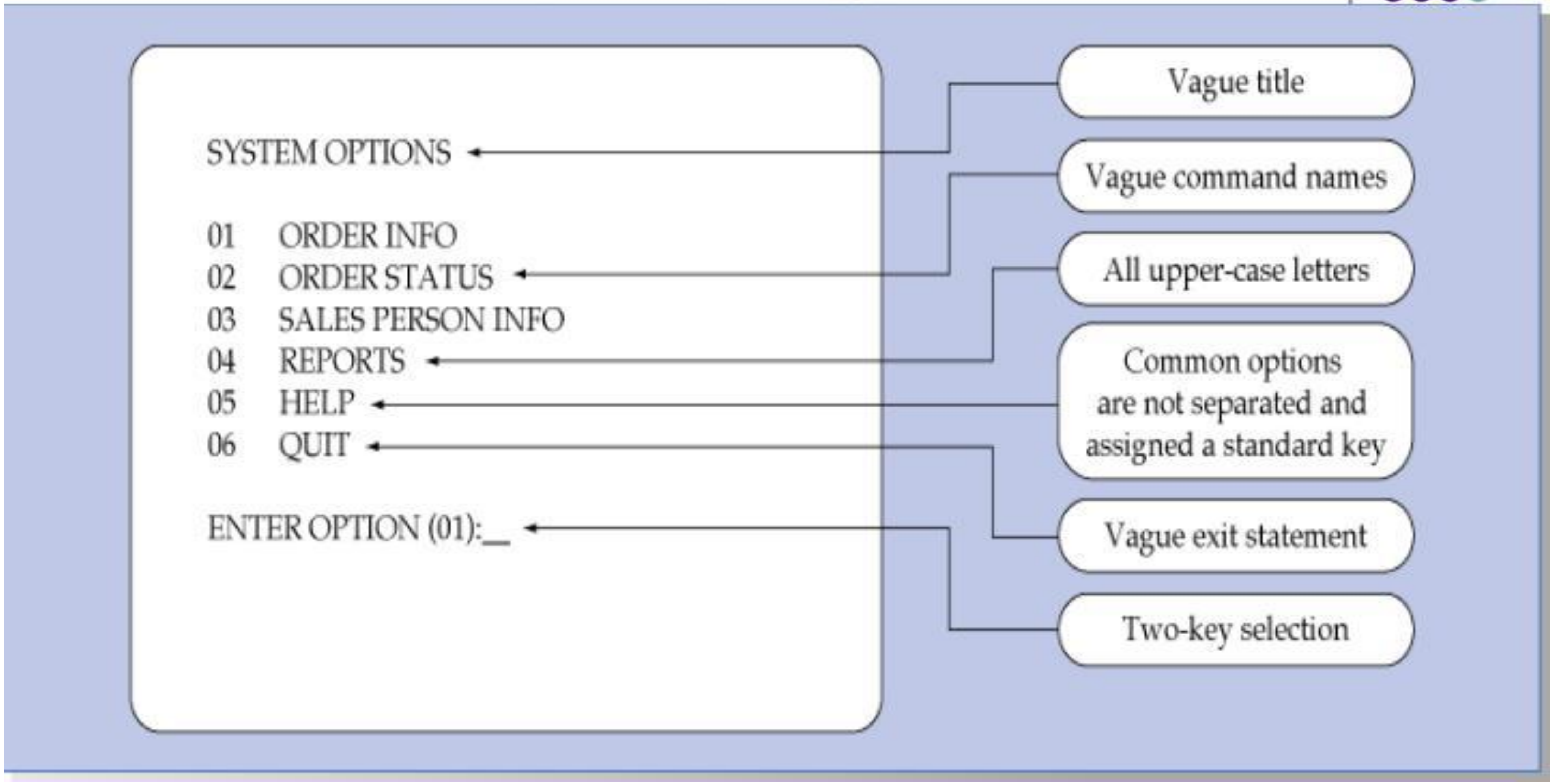
ii. Drop-Down Menu

- A menu positioning method that places the access point of the menu near the top-line of displays; when accessed, menus open by dropping down into display.
- Provide consistency and efficient display space
- Toolbars? Input Device Menus ?
- Most advanced operating environments, such as Microsoft windows or the Apple Macintosh, provide a combination of both pop-up and drop-down menu.

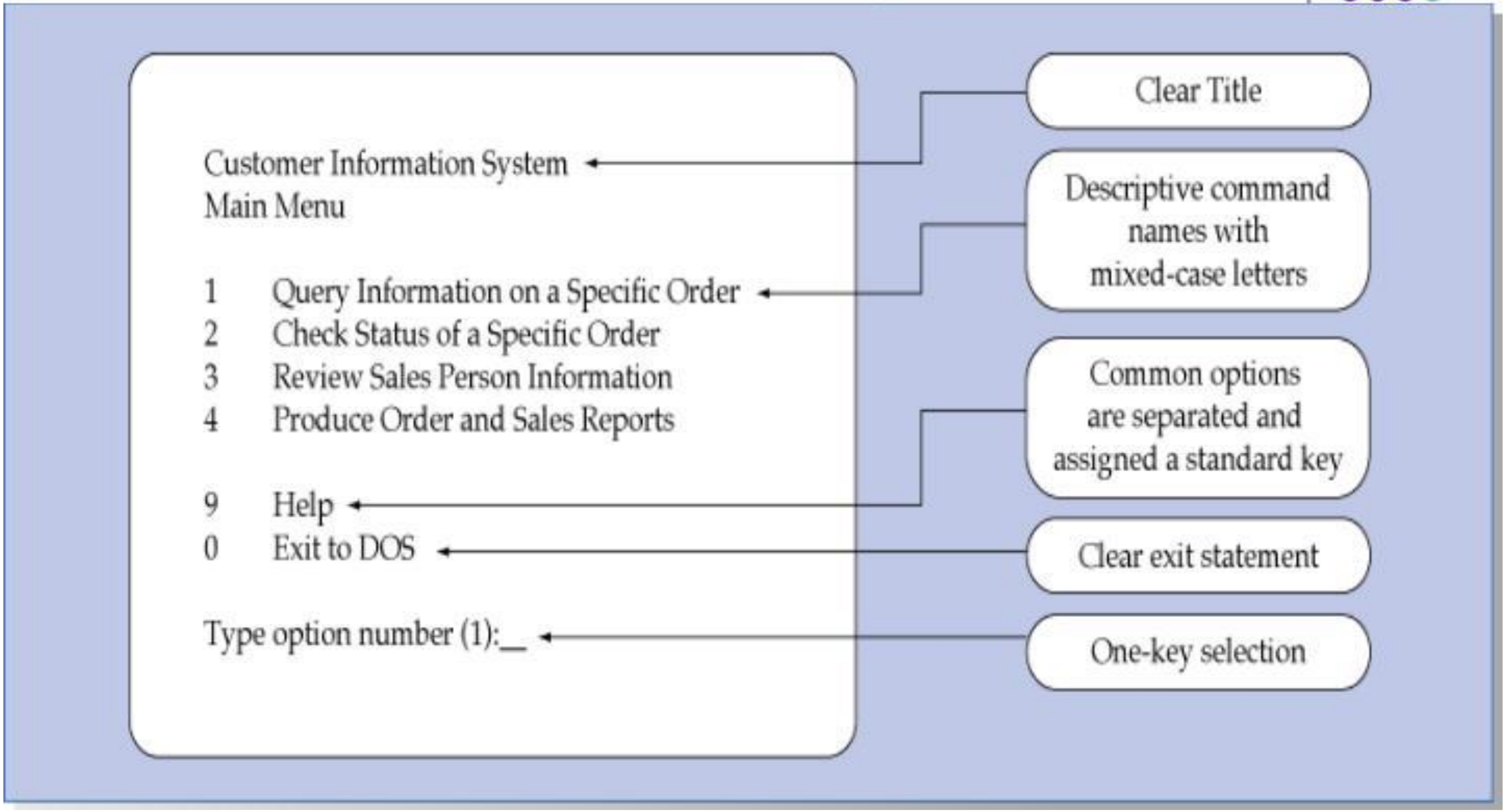
TABLE 14-1 Guidelines for Menu Design

Wording	• Each menu should have a meaningful title → Quit • Command verbs should clearly and specifically describe operations • Menu items should be displayed in mixed upper- and lower-case letters and have a clear, unambiguous interpretation
Organization	• A consistent organizing principle should be used that relates to the tasks the intended users perform; for example, related options should be grouped together and the same option should have the same wording and codes each time it appears
Length	• The number of menu choices should not exceed the length of the screen • Submenus should be used to break up exceedingly long menus
Selection	• Selection and entry methods should be consistent and reflect the size of the application and sophistication of the users • How the user is to select each option and the consequences of each option should be clear (e.g., whether another menu will appear)
Highlighting	• Highlighting should be minimized and used only to convey selected options...

Poor Menu Design



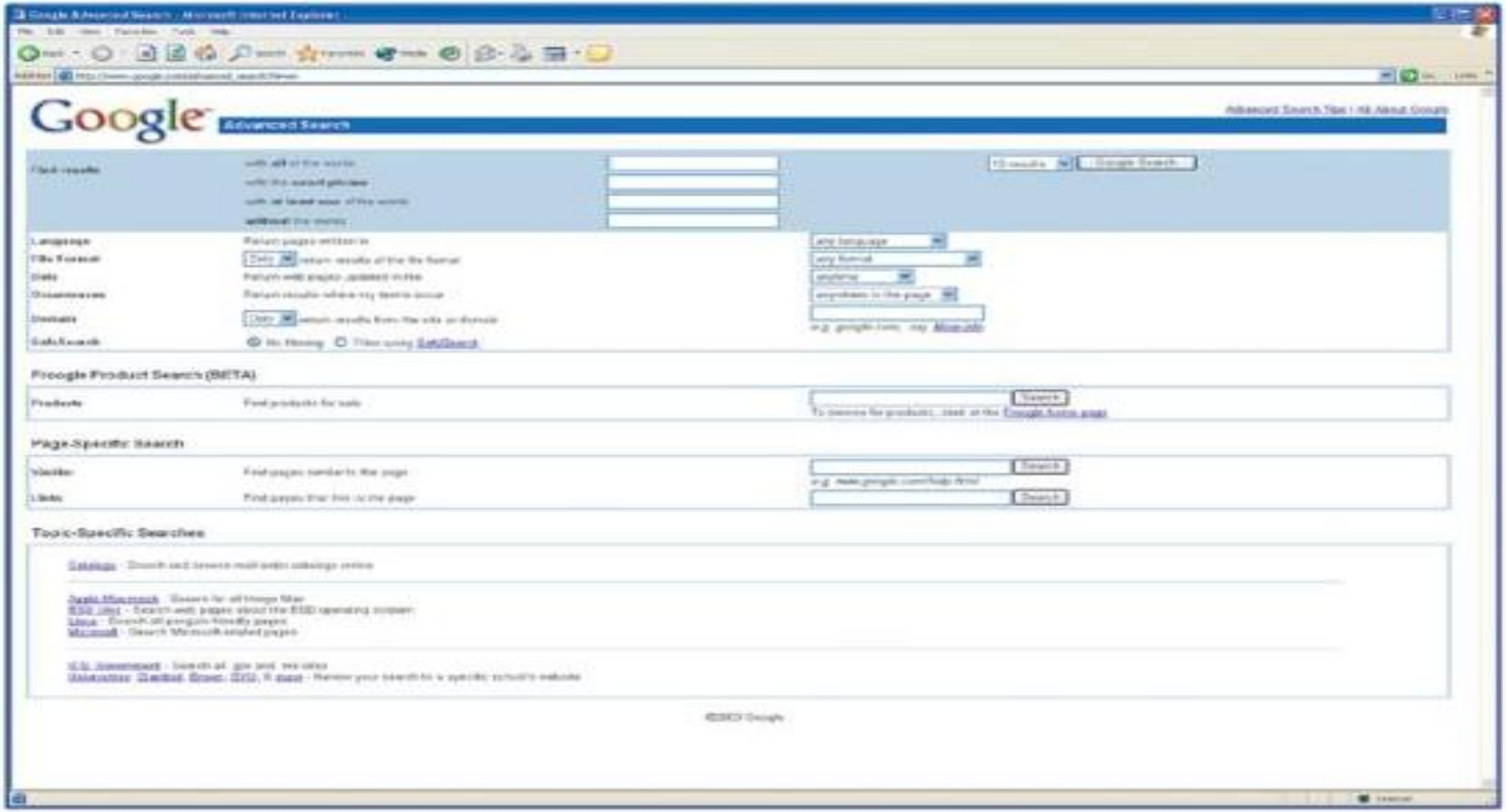
Good Menu Design



3. Form Interaction

- A highly intuitive human-computer interaction whereby data fields are formatted in a manner similar to paper-based forms.
- Allows users to fill in the blanks when working with a system.
- Measures of an effective design
- Self explanatory title and field headings
    - Fields organized into logical groupings
    - Distinctive boundaries
    - Default values
    - Displays appropriate field length
    - Minimizes the need to scroll windows

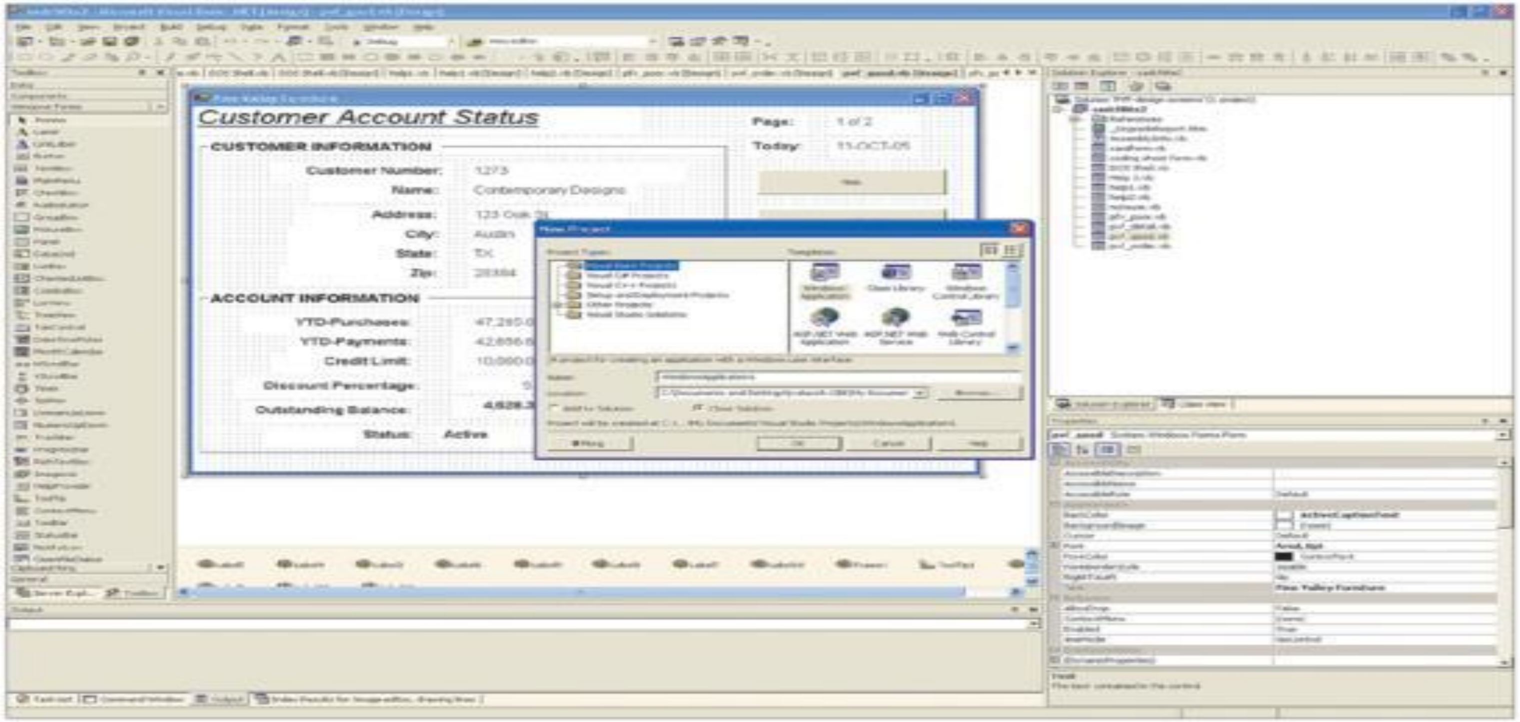
Figure 12-9 Example of form interaction from the Google Advanced Search Engine



4. Object Based Interaction

- A human computer interaction method in which symbols are used to represent commands or functions.
- Icons
  - Graphic symbols that look like the processing option they are meant to represent.
  - Use little screen space.
  - Can be easily understood by users.
- Used to display desktops in modern operating systems and starting interfaces

Figure 12-10 Object-based (icon) interface from Microsoft Visual Basic .NET



5. Natural Language Interaction

- **A human–computer interaction method whereby inputs to and outputs from a computer-based application are in a conventional spoken language such as English.**
- Not as viable (practical) as other interface methods.
- Can be tedious, frustrating, and time-consuming (might require learning).
- Used in narrow domains (database queries, accessibility).
- Might be applied using voice entry systems.

Hardware Options for System Interaction

Hardware Options for System Interaction

- Keyboard
- Mouse
- Trackball
- Joystick
- Touch Screen
- Light Pen
- Graphic Tablet
- Voice

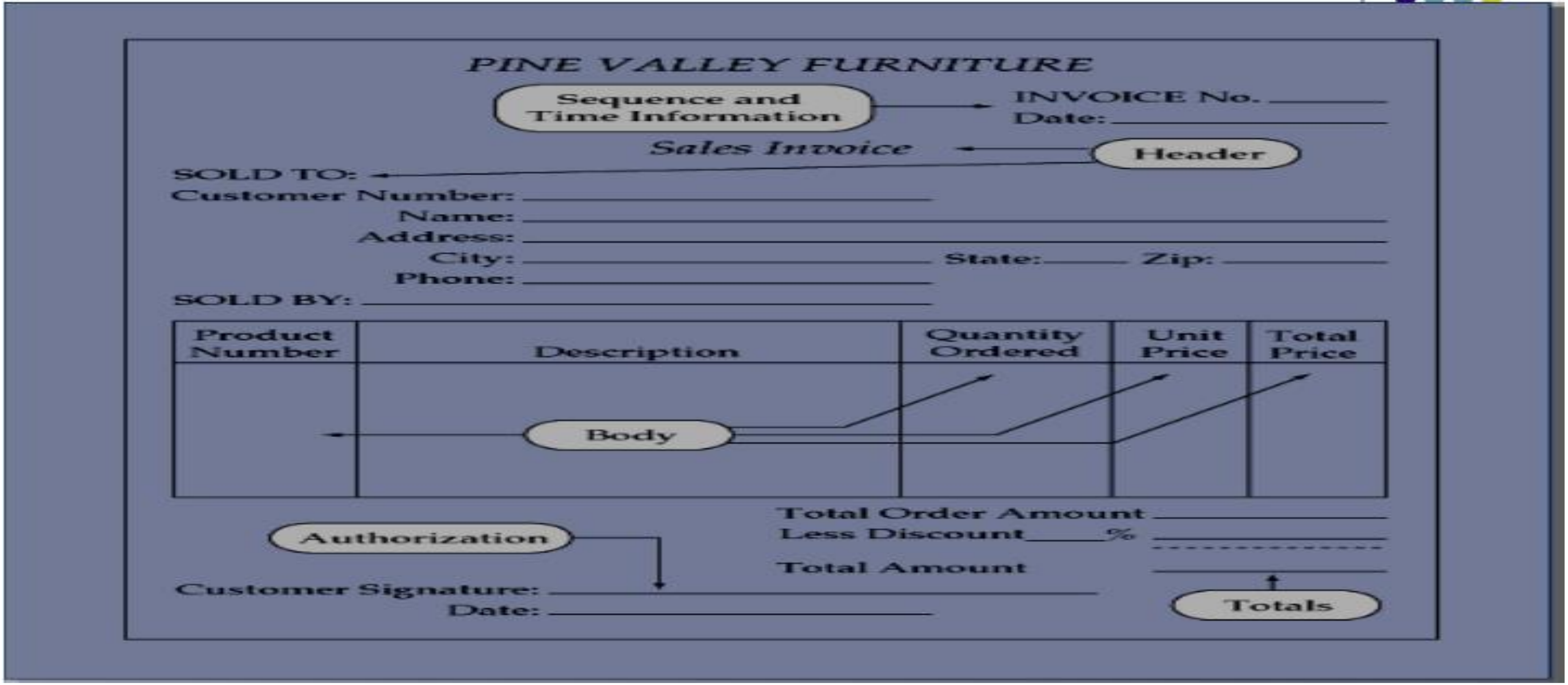
Designing Interfaces

- Guidelines for designing Interfaces are:
  1. **Designing Interface Layouts**
  2. **Structuring Data Entry Fields**
  3. **Controlling Data Input**
  4. **Providing Feedback**
  5. **Providing Help**

1. Designing Interface Layouts

- Use standard formats similar to both paper-based forms and reports.
- **Forms have several general common areas:**
  1. Header information
  2. Sequence and time-related information
  3. Instruction or formatting information
  4. Body or data details
  5. Totals or data summary
  6. Authorization or signatures
  7. Comment

- paper-based form for reporting customer sales activity.



computer-based activity form reporting customer sales activity.

Pine Valley Furniture

Customer Order Report

Today: 11-OCT-01

Order Number: 913-A36-01

CUSTOMER INFORMATION

Customer Number: 1273
Name: Contemporary Designs
Address: 123 Oak Street
City: Austin
State: TX
Zip: 28384

PRODUCT NUMBER	DESCRIPTION	QUANTITY ORDERED	UNIT PRICE	TOTAL PRICE
M128	Bookcase	4	200.00	800.00
B381	Cabinet	2	150.00	300.00
B210	Table	1	500.00	500.00
G200	Delux Chair	8	400.00	3,200.00

TOTAL ORDER AMOUNT	**4,800.00**
5% DISCOUNT	**240.00**
TOTAL AMOUNT DUE	**4,560.00**

Help | Print (password required) | Select Customer or Exit

- Some concern when designing the layout of computer-based forms is:
    - Screen navigation on data entry: screens should be left-to-right, top-to-bottom as on paper forms.
    - **Flexibility and consistency:**
        - Users should be able to move freely between fields.
        - Data should not be permanently saved until the user explicitly requests this.
        - Each key and command should be assigned to one function:
            - Cursor capabilities.
            - Editing capabilities.
            - Exit capabilities.
            - Help capabilities.

2. Structuring Data Entry Rules

Entry	Never require data that are already on-line or that can be computed (time,date,….)
Defaults	Always provide default values when appropriate (loan)
Units	Make clear the type of data units requested for entry (currency)
Replacement	Use character replacement when appropriate (lookup value/autocomplete)
Captioning	Always place a caption adjacent to fields
Format	Provide formatting examples(decimal points,.$,…)
Justify	Automatically justify data entries (text to left,numbers to right)
Help	Provide context-sensitive help when appropriate

3. **Controlling Data Input**

- One objective to interface design is to reduce data entry errors.
- Achieved by anticipating user errors and designing features into the system’s interfaces to avoid, detect and correct data entry mistakes
- Sources of data errors:
  - Appending: adding additional characters
  - Truncating: losing characters
  - Transcripting: entering invalid data into a field
  - Transposing: reversing the sequence of characters

4. **Providing Feedback**

- Feedback is very important in any conversation between people.
- Similarly, in a human computer interaction, system feedback is vital for satisfactory interaction.
- There are three types of system feedback
    1. **Status Information**
    2. **Prompting Cues**
    3. **Error and Warning Messages**

1. **Status Information**
   - Keeps users informed of what is going on in system.
   - Displaying status information is especially important if the operation takes longer than a second or two
   - Using different textual and visual effects: Progress bars, status bars and mouse icons
2. **Prompting cues**
   - Best to keep as specific as possible
     - ID: ………….
     - Enter Student ID: …………
3. **Error and Warning Messages**
   - Messages should be specific and free of error codes and jargon
   - User should be guided toward a result rather than scolded
   - Use terms familiar to the user
   - Be consistent in format and placement of messages

5. Providing Help

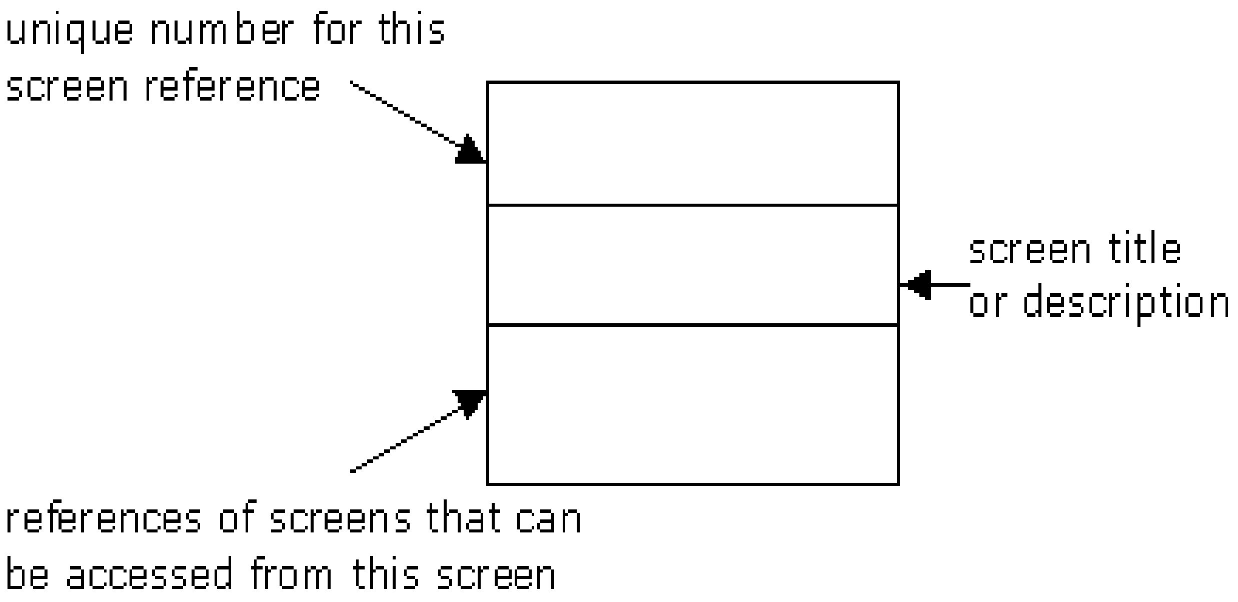
- Place yourself in users place when designing help guidelines
  - **Simplicity**
    - Help messages should be short and to the point
  - **Organization**
    - Information in help messages should be easily absorbed by users
  - **Demonstration**
    - It is useful to explicitly show users how to perform an operations (Tutorials, Animation)

Dialogue Design

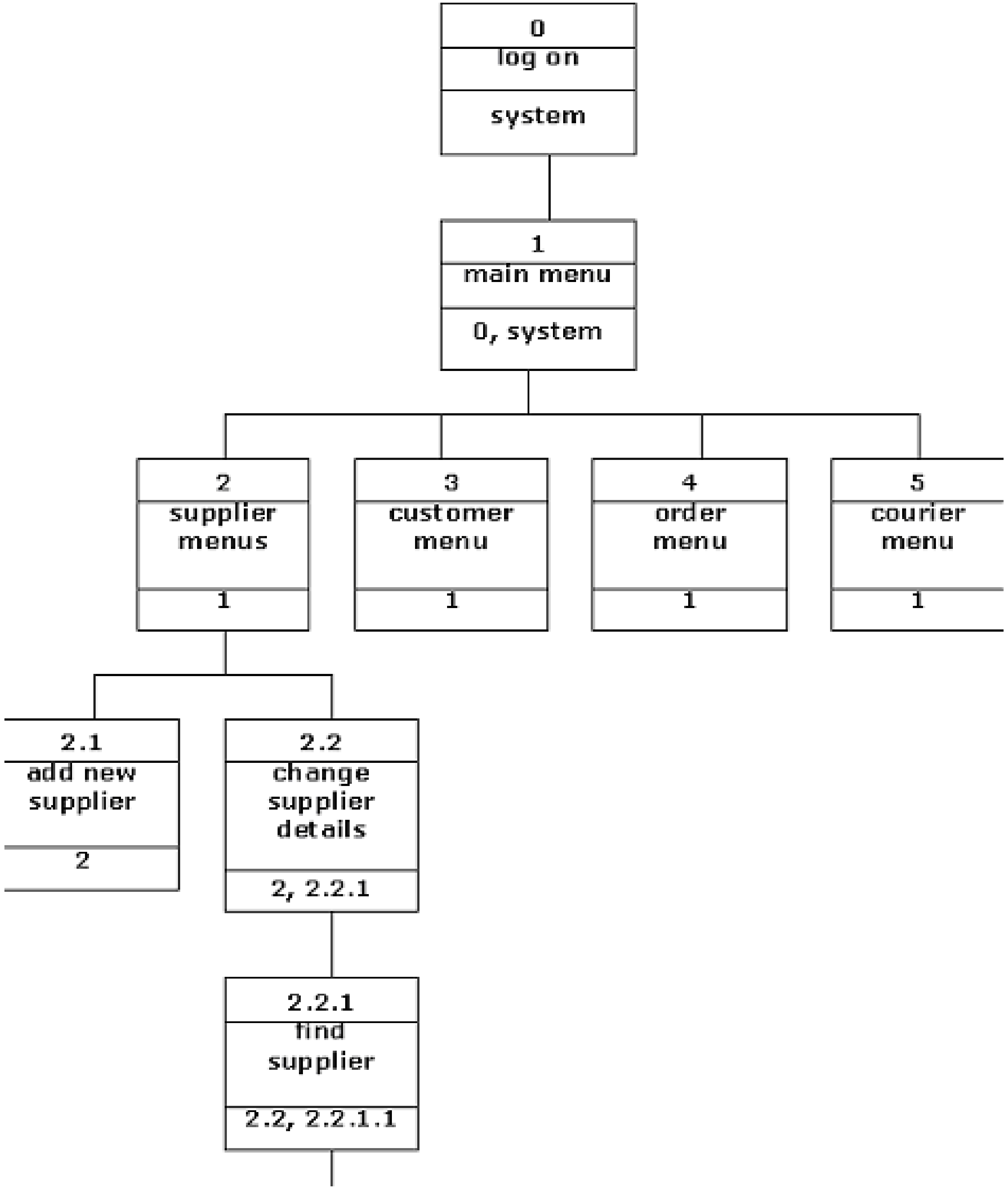
- Dialog is a written descriptions of flow of events
- A dialog is the construction of interaction between two or more beings or systems. a dialog is studied at three levels −
  - Dialog Design determines how the user interface will be created
  - To the user, the System Structure consists of the Menus
  - A Dialog can be expressed as the conversation between the user and the computer.

Dialogue Design

- A dialogue diagram uses a square divided into three parts. These squares are joined together by links without arrows. No arrows are needed as it is assumed that you can move forward and backward from screen to screen without limitations. Arrow heads would only be used if access to screens was one way only.



Example


0
log on
system
1
main menu
0, system
2
supplier menus
1
3
customer menu
1
4
order menu
1
5
courier menu
1
2.1
add new supplier
2
2.2
change supplier details
2, 2.2.1
2.2.1
find supplier
2.2, 2.2.1.1

Guidelines for Dialogue Design

- Consistency
- Shortcuts and Sequence
- Feedback
- Closure
- Error Handling
- Reversal
- Control
- Ease

Menu Layout

- Menus should be organized using task semantics.
- Positions should be shown by graphics, numbers or titles.
- Subtrees should use items as titles.
- Items should be grouped meaningfully.
- Items should be sequenced meaningfully.
- Brief items should be used.
- Consistent grammar, layout and technology should be used.
- Type ahead, jump ahead, or other shortcuts should be allowed.
- Jumps to previous and main menu should be allowed.
- Online help should be considered.

- The dialogue design process has three major steps:

1. **Designing the dialogue sequence**
2. **Building a prototype**
3. **Assessing Usability**

1. **Designing the dialogue sequence**
   - Define the dialogue sequence. How?
   - By having a clear understanding of the user, task, technological and environmental characteristics.
   - Transform these activities into Dialogue Diagram.

- **Dialogue Diagram:** A formal method for designing and representing human-computer dialogues using box and line diagrams

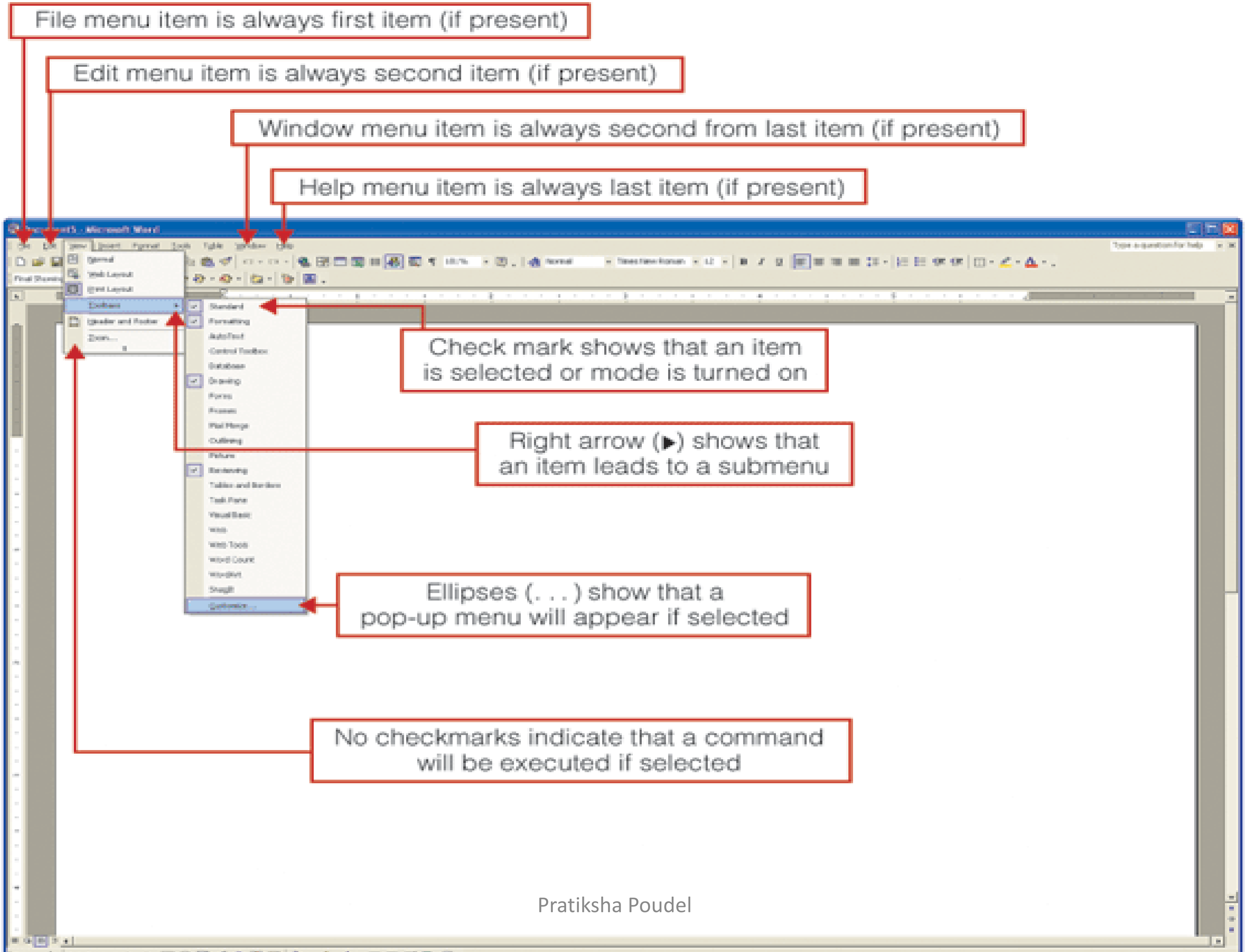
3. Assessing Usability

- Often optional activities
- Task is simplified by using graphical design environment.
- It includes:
  - Time to learn
  - Speed of performance
  - Rate of errors
  - Retention over time
  - Subjective satisfaction
- Assessing usability was discussed in the previous topic.

Designing Interfaces and Dialogues in Graphical Environments

- Become an expert user of the GUI environment.
  - Understand how other applications have been designed.
  - Understand standards.
- Gain an understanding of the available resources and how they can be used.
  - Become familiar with standards for menus and forms.

Figure 12-20 Highlighting graphical user interface design standards



GUI Window Properties That Can Be Turned On or Off

- Modality: require user to finish action before proceeding
- Resizable: allow user to change size of window
- Movable: allow user to reposition window
- Maximize: allow user to make window take entire screen
- Minimize: allow user to completely hide window
- System menu: allow window to have access to system level functions

GUI Dialogue Design Issues

- Goal is to establish the sequence of displays that users will encounter when working with system.
- Ability of some GUI environments to jump from application to application or screen to screen makes sequencing a challenge.
- One approach is to make users always resolve requests for information before proceeding.
- Dialogue diagramming helps analysts better manage the complexity of designing graphical interfaces.

Any Queries ???

Thank you!!!